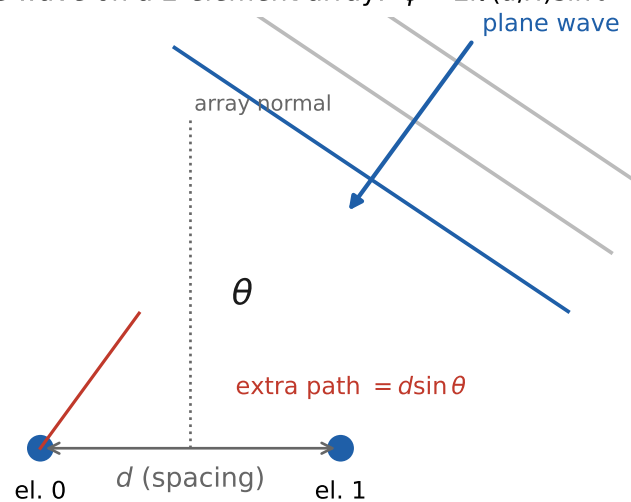


The SPF Signal Path

From Emission to Angle — a Mathematical Overview

transmission · propagation · down-conversion · phase difference · beamforming

Fig. 1 Plane wave on a 2-element array: $\phi = 2\pi(d/\lambda)\sin\theta$



$$\text{Core identity: } \phi = 2\pi \frac{d}{\lambda} \sin\theta = 2\pi f_c \frac{d\sin\theta}{c}$$

Signal Processing Fun (SPF) · generated from tutorials/build_signal_path_tutorial.py

1 Setup and notation

SPF locates a radio emitter by measuring, at a moving receiver, the phase difference between two closely-spaced antennas. One receiver = two antenna elements a distance d apart, sampled coherently. The whole chain reduces to one geometric fact (Fig. 1) carried faithfully through the radio hardware and the software stack.

f_c carrier (transmit) frequency; $\lambda = c/f_c$ is the wavelength

d physical spacing between the two antenna elements (metres)

d/λ spacing in wavelengths; production spans $\approx 0.12 - 1.55$ (above 0.5,

ϕ aliases -- see Section 8)

θ angle of arrival, measured from the array normal (broadside)

ϕ inter-element phase difference — the raw observable

f_{LO} receiver local-oscillator frequency (the 'base clock' of mixing)

Δf baseband offset $= f_c - f_{LO}$; SPF sets it to the 'intermediate' freq

f_s ADC sampling rate (SPF: 30 MHz on a 3 MHz signal)

Key idea, stated once: the inter-antenna phase ϕ is a carrier-frequency effect. It is set at RF, survives down-conversion unchanged, and is essentially independent of the sampling clock f_s . The sampling clock only has to satisfy Nyquist for the signal bandwidth.

2 Transmission

The emitter radiates a narrowband tone at carrier f_c . Written as a real passband signal and, equivalently, in complex analytic form:

$$s(t) = A \cos(2\pi f_c t + \psi) \quad \Leftrightarrow \quad \tilde{s}(t) = A e^{j(2\pi f_c t + \psi)}$$

A is amplitude, ψ an arbitrary start phase. The absolute phase ψ is unknowable and irrelevant — only differences will matter. In SPF the emitter is deliberately offset from the receiver LO by a known 'intermediate' frequency, so the tone lands at Δf in baseband (never at DC, which would collide with LO/DC leakage in the AD9361).

3 Propagation and array geometry

A plane wave arriving from angle θ reaches element 1 over an extra path length $\Delta d = d \sin \theta$ (Fig. 1). Extra path becomes extra phase at the carrier:

$$\phi = \frac{2\pi \Delta d}{\lambda} = \frac{2\pi d \sin \theta}{\lambda} = 2\pi \frac{d}{\lambda} \sin \theta$$

This is `theta_to_phi` in the code. Equivalently $\phi = 2\pi f_c \tau$ with $\tau = d \sin \theta / c$ the inter-element time delay. Note it is the CARRIER f_c that scales the phase.

Why phase, not time-of-arrival: with $d = 0.035$ m the delay is $\tau = d/c \approx 117$ ps. At $f_s = 30$ MHz one sample is 33 ns, so $\tau \approx 0.0035$ samples — unresolvable in time at ANY realistic ADC rate. But at $d/\lambda = 0.35$ the phase is $2\pi(0.35) \approx 126^\circ$: large and easy to read. Phased-array DF lives in the phase domain.

Because $\sin \theta = \sin(\pi - \theta)$, a single pair cannot separate θ from $\pi - \theta$: the front/back ambiguity (Fig. 3), resolved later by fusing two differently-oriented receivers.

Where the extra path $d \sin(\theta)$ comes from

Section 3 used $\phi = 2\pi(d/\lambda)\sin \theta$ without deriving it. It rests on two facts: a wave's phase advances with distance travelled, and geometry makes one antenna's path longer by $d \sin \theta$.

1. Phase advances 2π per wavelength

A travelling wave repeats every wavelength λ -- that is what wavelength MEANS. So moving an extra distance Δd along the propagation direction advances the phase by that fraction of a cycle:

$$\Delta\phi = 2\pi \frac{\Delta d}{\lambda}$$

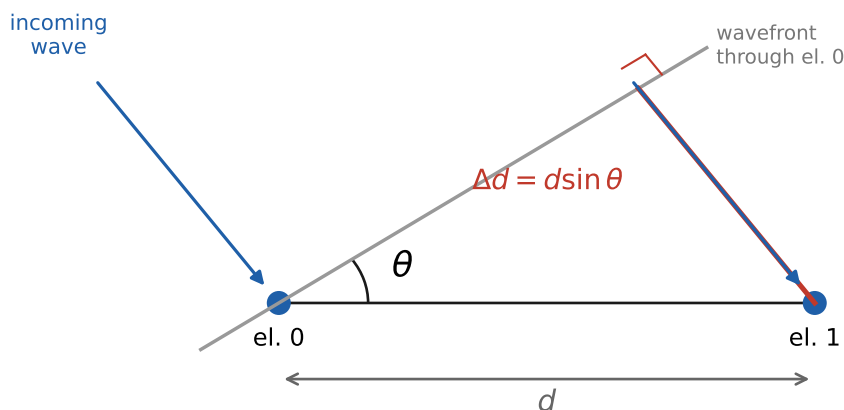
Two equivalent views. Spatial: a frozen snapshot of the wave is a sinusoid in space of period λ , so stepping Δd changes the cosine's argument by $(2\pi/\lambda)\Delta d$.

Temporal: the far antenna receives the wave later by $\tau = \Delta d/c$, and a carrier at f_c delayed by τ gains phase $2\pi f_c \tau = 2\pi \Delta d/\lambda$. Same answer -- extra distance, extra delay, extra phase.

2. Geometry: the extra path is $d \sin(\theta)$

The source is far, so wavefronts arrive flat and the two rays are parallel. Take the wavefront (a constant-phase line, perpendicular to the ray) that just touches element 0; the wave must travel an extra Δd to reach element 1 (Fig. 1b). In the right triangle with hypotenuse = baseline d and Δd opposite the angle θ :

Fig. 1b Extra path = projection of baseline d onto the ray = $d \sin \theta$



$$\sin \theta = \frac{\Delta d}{d} \Rightarrow \Delta d = d \sin \theta$$

(The triangle's angle is θ because the ray is θ off the normal, so the wavefront -- perpendicular to the ray -- is θ off the baseline.)

3. Combine

$$\phi = \frac{2\pi}{\lambda} \Delta d = 2\pi \frac{d}{\lambda} \sin \theta$$

- Broadside $\theta = 0$: $\sin \theta = 0$, $\Delta d = 0$, $\phi = 0$ -- the wavefront reaches both antennas at once (equidistant from the source).
- Endfire $\theta = 90^\circ$: $\sin \theta = 1$, $\Delta d = d$, ϕ is maximal -- the wave travels straight along the array axis.
- Hence $\sin \theta$ (zero at broadside, max at endfire); it would be $\cos \theta$ if θ were measured from the array axis, not the normal.

Example (SPF): $f_c = 2.4 \text{ GHz} \Rightarrow \lambda = 12.5 \text{ cm}$, $d = 3.5 \text{ cm}$ ($d/\lambda = 0.28$). At $\theta = 30^\circ$, $\Delta d = 1.75 \text{ cm} = 0.14\lambda \Rightarrow \phi \approx 50^\circ$; at $\theta = 90^\circ$, $\phi \approx 101^\circ$. A couple of cm of path becomes tens of degrees because it is a sizable fraction of the small wavelength.

The λ here is the CARRIER wavelength ($\lambda = c/f_c$). Δd is a fixed number of metres set by geometry; converting metres to phase divides by the carrier wavelength -- which is why the phase lives at the carrier, and a higher f_c gives more phase per degree of angle.

4 Receiving and down-conversion (high freq → low freq)

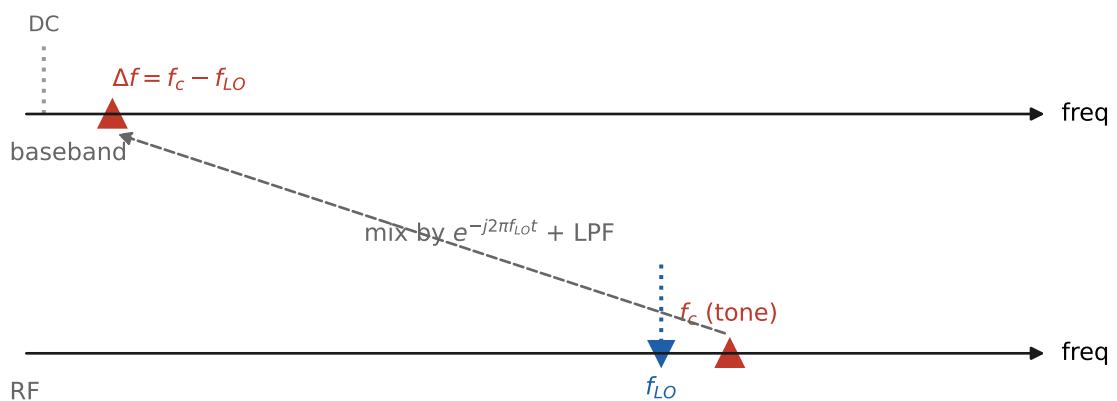
Each element $k \in \{0, 1\}$ sees the same tone plus its own spatial phase ϕ_k (with $\phi_0 = 0$, $\phi_1 = \phi$):

$$\tilde{s}_k(t) = A e^{j(2\pi f_c t + \psi + \phi_k)}$$

The receiver multiplies by the local oscillator $e^{-j2\pi f_{LO} t}$ (the same LO for both channels) and low-pass filters. Mixing shifts the spectrum down by f_{LO} (Fig. 2):

$$b_k(t) = \text{LPF}\{\tilde{s}_k(t) e^{-j2\pi f_{LO} t}\} = A e^{j(2\pi \Delta f t + \psi + \phi_k)}, \quad \Delta f = f_c - f_{LO}$$

Fig. 2 Down-conversion shifts $f_c \rightarrow \Delta f$; the spatial phase ϕ rides through unchanged



This is the crux the question asks about: the down-conversion moves the signal from f_c to Δf , but the spatial phase ϕ_k appears as an ADDITIVE CONSTANT on the baseband phasor — identical before and after the mixer. The mixer changes the frequency, not the spatial phase. So ϕ is 'placed' as a fixed per-channel offset that is simply carried along into baseband.

5 The phase difference through the stack

Form the phase difference of the two baseband channels (this is `get_phase_diff` in `rf.py`) :

$$\Delta\phi(t) = \angle b_1(t) - \angle b_0(t) = [2\pi\Delta ft + \psi + \phi_1] - [2\pi\Delta ft + \psi + \phi_0] = \phi$$

Everything common to both channels cancels at every sample: the LO/baseband rotation $2\pi\Delta ft$, the source phase ψ , and the unknown absolute offset. Only the geometry-bearing ϕ survives. This is why (a) the intermediate offset Δf is harmless — it cancels sample-by-sample; and (b) absolute phase is discarded — the pipeline only ever needs the DIFFERENCE.

SPF then averages $\Delta\phi$ over a window with a trimmed CIRCULAR mean (angles wrap at $\pm\pi$) , giving a robust per-window phase and a spread (stddev) used as a confidence weight.

Sign convention in the implementation: the code computes the OPPOSITE difference -- `get_phase_diff` (`rf.py:745`) returns $\angle b_0 - \angle b_1 = -\phi$ -- and the dataset's ground-truth target carries a matching $-\sin\theta$ (`spf_dataset.py:1513`), so the two minus signs cancel end-to-end. Which physical antenna is element 0 is set by cabling, not code; the pair (measurement sign, target sign) is what must stay consistent, and every filter likewise negates the spacing.

The one term that does NOT cancel is a per-channel LO phase mismatch between the two RX chains. It survives as a fixed offset — the `phi_drift / AD9361 RX1-RX2` phase inversion — which SPF removes with a calibration step (and models with opposite sign on the two receivers in the synthetic generator).

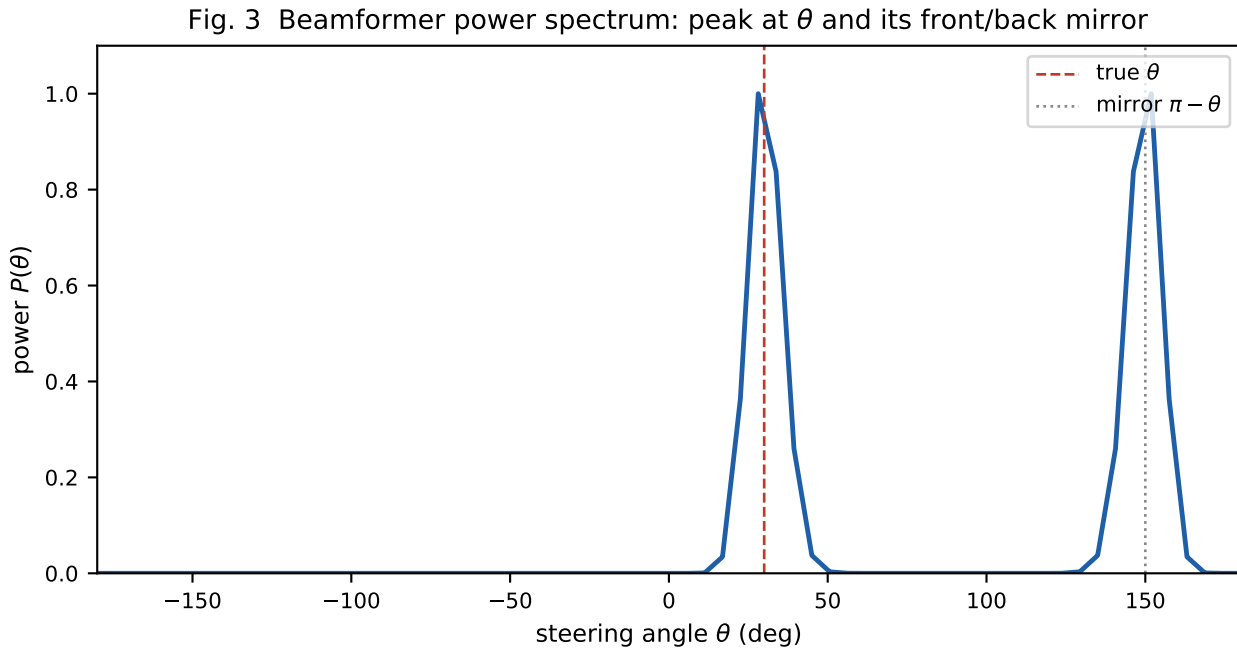
6 From phase to angle: beamforming

Rather than invert $\phi \rightarrow \theta$ directly (which is 2-to-1), SPF SCANS candidate angles. For each θ_i build a steering vector that predicts the phase a wave from θ_i would imprint, apply it to the signal, and measure output power:

$$a(\theta_i) = e^{-j2\pi(d/\lambda)\sin\theta_i}, \quad P(\theta_i) = |a(\theta_i) \cdot x|$$

(No conjugate is applied at multiply time -- the compensating minus sign is already baked into the e^{-j} steering definition; the code does a plain product, `rf.py:1266`. And in the implementation the steering phase is split $\pm\phi/2$ across the two elements at $\pm d/2$ rather than all on element 1 -- an equivalent global-phase choice, `rf.py:1020-1023`.)

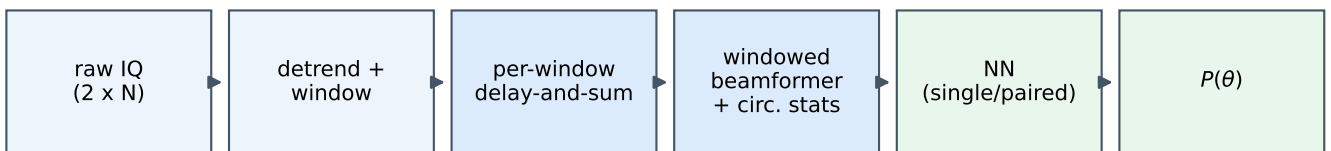
$P(\theta)$ peaks where the steering phase matches the measured phase (Fig. 3). With a 2-element array the response is bimodal — a peak at the true θ and at its mirror $\pi - \theta$. SPF evaluates this on a grid of $n_\theta = 65$ angles (linspace includes both $\pm\pi$ endpoints, which are the same physical angle).



7 How it flows through the SPF stack

The raw IQ is processed in windows: detrend, then a per-window delay-and-sum beamformer produces a $(n_{\text{windows}} \times 65)$ power map (the `windowed_beamformer`), alongside per-window circular phase mean/stddev (`all_windows_stats`). These features — NOT the raw IQ — feed the neural network, which outputs a distribution over the 65 angle bins.

Fig. 4 The SPF processing stack (stage-1 path)



Symmetry is handled deliberately: the network is trained invariant to left/right ($\theta \rightarrow -\theta$, data augmentation) and its target is folded front/back ($\theta \rightarrow \text{sign}(\theta)\pi - \theta$), because a single pair genuinely cannot resolve either. A second, differently-oriented receiver (the paired model) breaks the ambiguity.

8 Hardware artifacts and real SPF numbers

The idealized $b_k(t)$ picks up real-world perturbations. Those SPF models explicitly (and why):

- thermal / phase noise $\Rightarrow$ Gaussian on ϕ (broadens the peak)
- unknown absolute phase $\Rightarrow$ random ψ on both elements (cancels)
- inter-radio LO offset $\Rightarrow$ fixed ϕ_{drift} (calibrated)
- intermittent emitter $\Rightarrow$ noise windows (what segmentation detects)

Not modeled (levers for realism): multipath, Doppler, IQ imbalance, DC offset, ADC quantization, AGC dynamics, real packet modulation.

Representative receiver parameters

carrier $f_c \rightarrow 2.4$ or 5.766 GHz

wavelength $\lambda = c/f_c \rightarrow \approx 12.5 / 5.2$ cm

spacing d / ratio $d/\lambda \rightarrow 0.025 - 0.08$ m / $0.12 - 1.55$
(> 0.5 aliased for many sets)

sample rate $f_s \rightarrow 30$ MHz (signal BW ≈ 3 MHz)

inter-element delay $\tau_{\text{max}} \rightarrow \approx 117$ ps ($\ll$ one sample)

angle grid $n_\theta \rightarrow 65$ bins on $[-\pi, \pi]$

9 One-page recap

Geometry sets the observable at the carrier:

$$\phi = 2\pi(d/\lambda)\sin\theta$$

Down-conversion by a shared LO moves $f_c \rightarrow \Delta f$ but carries ϕ through as an additive constant:

$$b_k(t) = A e^{j(2\pi\Delta f t + \psi + \phi_k)}$$

The inter-channel difference cancels all common terms and recovers pure geometry:

$$\angle b_1 - \angle b_0 = \phi \Rightarrow \theta \text{ (up to front/back)}$$

Sampling clock: needs only $f_s \geq$ signal bandwidth (Nyquist) plus channel COHERENCE (shared LO + clock). It does not set, and is not set by, the phase ϕ .

Appendix A IQ modulation: why the samples are complex

Every stage above treats each baseband sample as one complex number. That number is produced by IQ (in-phase / quadrature) demodulation, and it is the reason SPF can read phase at all.

The idea

Any narrowband passband signal is a slowly-varying complex envelope $z(t) = I(t) + jQ(t)$ riding on the carrier:

$$s(t) = I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t) = \text{Re}\{ z(t) e^{j2\pi f_c t} \}$$

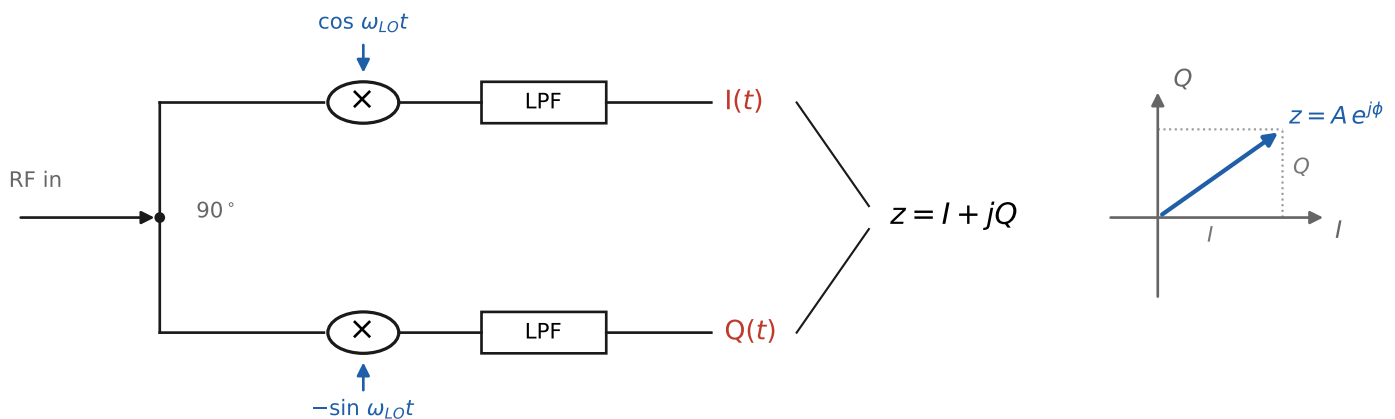
The two real branches carry the whole story -- $I = A\cos\phi$ (in-phase), $Q = A\sin\phi$ (quadrature) -- so amplitude and phase fall straight out:

$$|z| = \sqrt{I^2 + Q^2} = A, \quad \angle z = \arctan(Q/I) = \phi$$

A receiver recovers I and Q by mixing the antenna voltage with two copies of the local oscillator 90° apart ($\cos$ and $-\sin$), then low-pass filtering each branch (Fig. 5).

The AD9361 in the Pluto does exactly this and hands software a stream of complex samples -- SPF's `signal_matrix` is `complex64`.

Fig. 5 Quadrature (IQ) demodulator: two LO copies 90° apart give the complex sample $z = I + jQ = A e^{j\phi}$



Why use it -- hardware

- Half the ADC rate: a complex sample at f_s spans a full bandwidth f_s ($-f_s/2$ to $+f_s/2$); a real ADC needs $2f_s$ for the same span.
- Zero-/low-IF front end: mixing straight to baseband removes the high-frequency IF stage and its filters -- what lets an SDR fit on one chip.
- Image rejection for free: a real mixer folds the signal and its mirror image together; the 90° I/Q relationship separates $+$ and $-$ frequencies and cancels the image, no dedicated filter needed.
- Amplitude AND phase in one shot: both ADC channels together deliver the full complex envelope every sample -- no phase reconstruction.

Why use it -- signal processing

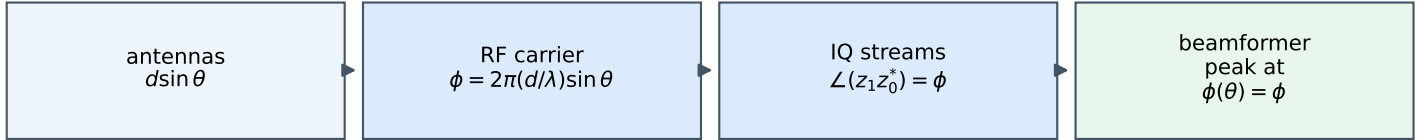
- Instantaneous phase is free: $\phi = \arctan(Q/I)$. SPF's core operation -- the inter-antenna difference $\angle z_1 - \angle z_0$ (get_phase_diff) -- exists only because samples are complex.
- Positive and negative frequencies stay distinct (one-sided spectrum): a tone above the LO is told from one below -- what makes the intermediate offset Δf unambiguous.
- Frequency shifts and beamforming are just complex multiplies: down-conversion is $\times e^{-j\omega t}$; a steering vector is $e^{-j\phi(\theta)}$ applied to the complex signal (exactly `rf.py` steering vectors $\cdot$ signal matrix).
- Common-mode rotations cancel cleanly: the LO phase and the $\Delta f t$ rotation are shared multiplicative phases that drop out of the difference (the cancellation in Section 5) -- natural arithmetic in the complex plane.

Without IQ you would sample a real passband signal: 2x the ADC rate, an image-reject filter, and a Hilbert transform to recover instantaneous phase -- and you still could not tell $+\Delta f$ from $-\Delta f$. IQ is what makes phase-difference direction-finding practical.

Appendix B From spacing d to the beamformer peak

This appendix follows one quantity -- the inter-antenna phase difference ϕ -- from the physical spacing d to where it becomes a peak in the beamformer. It is the SAME ϕ at every stage; only its representation changes (Fig. 6).

Fig. 6 One phase difference, four representations



the same ϕ at every stage -- only its representation changes

1. At the antennas (RF)

A wave from angle θ reaches element 1 over an extra path $d \sin \theta$ -- an extra carrier phase

$$\phi = 2\pi(d/\lambda)\sin \theta$$

so the two antenna voltages differ by exactly ϕ :

$$s_0 = A e^{j(2\pi f_c t + \psi)}, \quad s_1 = A e^{j(2\pi f_c t + \psi + \phi)}$$

The spacing d enters once, here, at the carrier.

2. Into the IQ samples (down-conversion)

Both channels are mixed by the SAME LO $e^{-j2\pi f_{LO} t}$ and low-pass filtered. The frequency drops to $\Delta f = f_c - f_{LO}$; the spatial phase ϕ is untouched:

$$z_0 = A e^{j(2\pi \Delta f t + \psi)}, \quad z_1 = A e^{j(2\pi \Delta f t + \psi + \phi)}$$

In the complex IQ stream ϕ is the constant angle by which z_1 leads z_0 -- equivalently the phase of their cross-product, where the common LO/rotation terms cancel:

$$z_1 z_0^* = A^2 e^{j\phi} \Rightarrow \phi = \angle(z_1 z_0^*)$$

So d has trickled from a carrier phase into a fixed rotation between two complex sample streams.

3. Reading phi optimally

To recover ϕ from noisy samples, combine COHERENTLY over the window: the maximum-likelihood estimate is the angle of the summed cross-product,

$$\hat{\phi} = \angle \left(\sum_t z_1(t) z_0^*(t) \right)$$

Summing the complex products BEFORE taking the angle amplitude-weights each sample by its SNR -- the optimal translation of antenna phase into a number. SPF's per-sample `get_phase_diff` + trimmed circular mean is a robust near-equivalent.

4. Into the beamformer (where phi shows up)

The beamformer scans candidate angles. For θ_i it applies the modeled steering phase $\phi(\theta_i) = 2\pi(d/\lambda)\sin\theta_i$ to element 1 and coherently sums the two elements:

$$y(\theta_i) = z_0 + e^{-j\phi(\theta_i)} z_1, \quad P(\theta_i) = |y(\theta_i)|$$

Substituting $z_1 = z_0 e^{i\phi}$ from stage 2:

$$P(\theta_i) = |z_0| |1 + e^{j(\phi - \phi(\theta_i))}| = 2 |z_0| \left| \cos\left(\frac{\phi - \phi(\theta_i)}{2}\right) \right|$$

(The code returns this magnitude, not its square -- `beamformer_given_steering_nomean`, `rf.py:1302-1306`; the argmax over θ_i is identical either way.) The response is maximal exactly when the modeled phase matches the measured one, $\phi(\theta_i) = \phi$ (i.e. $\sin\theta_i = \sin\theta$). THAT is where ϕ appears: as the LOCATION of the beamformer peak. Crucially the same d sits in both the physical ϕ and the steering $\phi(\theta_i)$, so they must use the identical spacing (`rx_wavelength_spacing`) or the peak lands at a biased angle.

One quantity, four faces: an extra path (metres) -> a carrier phase (RF) -> a fixed rotation between two IQ streams (baseband) -> the argmax of a power spectrum (beamformer). 'Optimal' at each step means coherent summation -- exactly what both the ML phase estimate and the beamformer do.